

☒ EASY (10 Questions)

1. What is the maximum number of unique addresses in IPv4?

- A. 128
- B. 1 million
- C. ~4.3 billion ☒
- D. ~340 undecillion

☒ Answer: C

Explanation: IPv4 uses a 32-bit address space, giving approximately 4.3 billion addresses (2^{32}).

2. Which protocol does IPv6 support natively that IPv4 does not?

- A. ARP
- B. NAT
- C. IPsec ☒
- D. DHCP

☒ **Answer: C**

Explanation: IPv6 was designed with native support for IPsec, providing better built-in security.

3. DFS stands for:

- A. Dynamic File System
- B. Distributed File System ☒
- C. Dual Folder Share
- D. Distributed Folder Sync

☒ **Answer: B**

Explanation: DFS allows you to create a unified namespace for accessing shared folders from multiple servers.

4. Which of the following is a valid IPv6 address?

- A. 192.168.1.1
- B. 2001:0db8:85a3::8a2e:0370:7334 ☒
- C. 256.256.256.256

D. fe80-0000-0000-0000

☒ **Answer:** B

Explanation: This is a properly formatted IPv6 address using colons and optional shorthand (::).

5. Which type of domain controller is recommended for a branch office with limited physical security?

- A. PDC
- B. ADC
- C. RODC ☒
- D. FSMO

☒ **Answer:** C

Explanation: A Read-Only Domain Controller (RODC) offers authentication and caching without full replication, ideal for insecure locations.

6. The main benefit of DFS Namespaces is:

- A. Speed
- B. Logical file access ☒
- C. Backup
- D. Encryption

☒ **Answer:** B

Explanation: DFS Namespace creates a unified structure for users to access files across multiple servers.

7. Which version of IP uses 128-bit addresses?

- A. IPv4
- B. IPv5
- C. IPv6 ☒
- D. None

☒ **Answer:** C

Explanation: IPv6 uses 128-bit addresses, supporting a vastly larger address pool.

8. Which command shows current IP configuration in Windows?

- A. netstat
- B. ipconfig
- C. ping
- D. arp

☒ **Answer:** B

Explanation: ipconfig displays IP configuration on Windows machines.

9. DFS Replication uses:

- A. Master-slave model
- B. Multi-master replication
- C. Asymmetric encryption
- D. Broadcast packets

☒ **Answer:** B

Explanation: DFSR uses multi-master replication, allowing all members to send updates.

10. BranchCache helps to:

- A. Encrypt WAN data
- B. Cache web pages
- C. Cache remote content locally ☒
- D. Store backup images

☒ **Answer: C**

Explanation: BranchCache stores frequently accessed content locally to save WAN bandwidth.

☒ MEDIUM (20 Questions)

11. In DFS, the logical namespace starts with:

- A. C:\DFS
- B. \DFS
- C. \domainname\dfsroot ☒
- D. /root/dfs

☒ **Answer:** C

Explanation: The DFS namespace uses UNC paths like \\domain.local\dfsroot.

12. What does an RODC restrict compared to a standard Domain Controller?

- A. DNS functionality
- B. File sharing
- C. Writing changes to Active Directory ☒
- D. Kerberos authentication

☒ **Answer:** C

Explanation: RODCs cannot make changes to Active Directory—they only cache read-only data.

13. Which IPv6 address type is used for a single interface on a single host?

- A. Multicast
- B. Global Unicast

- C. Anycast
- D. Link-local ☒

☒ **Answer:** D

Explanation: Link-local addresses are used for communication within a local link.

14. What command enables IPv6 on a Windows interface?

- A. ipv6 enable
- B. netsh interface ipv6 set interface INTERFACENAME enabled ☒
- C. set ipv6=1
- D. enable-ipv6

☒ **Answer:** B

Explanation: You use netsh to configure advanced interface settings like IPv6.

15. Which statement about DFSR is true?

- A. It supports only one master
- B. It replicates changes using a polling method
- C. It uses Remote Differential Compression ☒
- D. It copies all files fully each time

☒ **Answer: C**

Explanation: DFSR uses Remote Differential Compression (RDC) to minimize data transfer.

16. Branch office solutions typically use which combination for security and availability?

- A. NAT and DNS
- B. RODC and BranchCache ☒
- C. RADIUS and DHCP
- D. VPN and NAT

☒ **Answer: B**

Explanation: RODC allows secure read-only domain access; BranchCache improves performance by caching.

17. Which command shows the route taken by packets?

- A. ping
- B. netstat
- C. tracert ☒
- D. arp

☒ **Answer:** C

Explanation: tracert (Windows) shows each hop along the packet path.

18. What must be configured before using DFS replication?

- A. Group Policy
- B. Firewall rules
- C. Shared folders and replication groups ☒
- D. Active Directory Certificate Services

☒ **Answer:** C

Explanation: DFSR needs shared folders added to replication groups.

19. What protocol does IPv6 use for address resolution?

- A. ARP
- B. ICMPv6 ☒
- C. IGMP
- D. TCP

☒ **Answer:** B

Explanation: IPv6 uses ICMPv6 with NDP instead of ARP for address resolution.

20. DFS Replication uses which of the following methods for detecting changes?

- A. File timestamps
- B. USN journal ☒
- C. Ping
- D. Polling

☒ **Answer:** B

Explanation: DFSR uses the USN journal to efficiently track file changes.

21. A DFS Namespace server must be:

- A. A domain controller
- B. A DHCP server
- C. A member of the domain ☒
- D. Running Windows 7

☒ **Answer:** C

Explanation: It must be part of the domain to use domain-based DFS.

22. Which type of IPv6 address starts with fe80::?

- A. Multicast
- B. Global Unicast
- C. Link-local ☒
- D. Anycast

☒ **Answer:** C

Explanation: fe80::/10 is reserved for link-local addresses.

23. What is the primary function of DHCP in a branch office?

- A. Provide remote access
- B. Assign IP addresses automatically ☒
- C. Store logs
- D. Secure DNS queries

☒ **Answer:** B

Explanation: DHCP auto-assigns IPs to devices in any network segment.

24. What tool manages DFS configurations in Windows Server?

- A. Disk Management
- B. DNS Manager
- C. DFS Management ☒

D. GPO Editor

☒ **Answer: C**

Explanation: DFS Management is a dedicated GUI for DFS Namespaces and Replication.

25. What does an RODC not hold by default?

- A. DNS zones
- B. Passwords ☒
- C. Group Policy
- D. LDAP schema

☒ **Answer: B**

Explanation: By default, RODCs do not cache passwords unless explicitly configured.

26. Which command tests both IPv4 and IPv6 connectivity?

- A. ping

- B. netstat
- C. nslookup
- D. ping -6 ☒

☒ **Answer:** D

Explanation: ping -6 sends ICMP echo requests over IPv6.

27. What limits the replication bandwidth in DFSR?

- A. Time
- B. File size
- C. Schedule and bandwidth throttling ☒
- D. User permissions

☒ **Answer:** C

Explanation: DFSR can be scheduled and throttled to avoid WAN congestion.

28. How does BranchCache operate in Hosted Cache mode?

- A. Each client stores its own cache
- B. Content is stored centrally ☒
- C. Uses DFS
- D. Relies on Active Directory

☒ **Answer:** B

Explanation: In Hosted Cache mode, a central server in the branch holds the cache.

29. What is needed to support IPv6 routing?

- A. ARP
- B. NAT
- C. Router Advertisement ☒
- D. RDP

☒ **Answer:** C

Explanation: Routers use Router Advertisements (RA) to help configure IPv6 hosts.

30. DFS replication is ideal for:

- A. Real-time transactions
- B. Synchronizing logins
- C. Replicating static file shares across sites ☒
- D. DNS zone transfers

☒ **Answer:** C

Explanation: DFSR is used for replicating content like documents and files across servers.

☒ **HARD (10 Questions)**

31. How does DFSR avoid conflict when two users change the same file?

- A. Drops both copies
- B. Locks the file globally
- C. Keeps the latest file and moves the other to ConflictAndDeleted folder ☒
- D. Merges changes automatically

☒ **Answer: C**

Explanation: DFSR moves one copy to the ConflictAndDeleted folder to preserve changes.

32. Which mechanism ensures IPv6 hosts configure themselves automatically?

- A. DHCPv6 only
- B. NDP and SLAAC ☒
- C. DNS suffix
- D. NAT64

☒ **Answer: B**

Explanation: SLAAC uses Router Advertisements to let hosts auto-configure.

33. What DFSR feature avoids replicating unchanged parts of files?

- A. USN polling
- B. Remote File Comparison
- C. Remote Differential Compression ☒

D. File Syncing

☒ **Answer: C**

Explanation: RDC compares blocks and only transfers changed parts.

34. Which RODC feature prevents replication of sensitive groups?

- A. Delegation tab
- B. Password replication policy ☒
- C. Global catalog
- D. DNS scavenging

☒ **Answer: B**

Explanation: PRP controls which credentials can be cached on an RODC.

35. IPv6 supports dual-stack operation by:

- A. Running only IPv6

- B. Disabling NAT
- C. Running IPv4 and IPv6 together ☒
- D. Bridging networks

☒ **Answer: C**

Explanation: Dual-stack allows both IPv4 and IPv6 to operate on the same machine.

36. BranchCache Hosted Cache mode requires:

- A. NAT
- B. WINS
- C. A designated cache server ☒
- D. DFS Namespace

☒ **Answer: C**

Explanation: Hosted mode uses a single server to cache content for the branch.

37. In DFS, a referral is:

- A. A shortcut
- B. A DNS record
- C. A list of servers for a folder target ☒
- D. A user alias

☒ **Answer:** C

Explanation: DFS referrals guide clients to actual folder targets.

38. Which command clears IPv6 settings?

- A. netsh interface ipv6 reset ☒
- B. ipconfig /renew
- C. route delete
- D. disable-ipv6

☒ **Answer:** A

Explanation: This resets all IPv6 config on the interface.

39. What DNS record is essential for DFS Namespace in domain mode?

- A. A record
- B. PTR
- C. SRV ☒
- D. TXT

☒ **Answer:** C

Explanation: DFS in domain mode depends on SRV records in AD-integrated DNS.

40. Which setting prevents replication of stale files?

- A. ConflictAndDeletedQuota
- B. FileScreening
- C. Staging Quota
- D. Staging Cleanup Policy ☒

☒ **Answer:** D

Explanation: This ensures old staging files are removed and not replicated.